Table 1: Standardized computational time for the core PU configurations. Entries are median end-to-end fitting times with interquartile ranges over five fixed data realizations. Timing includes the candidate hyperparameter search, validation-based model selection, and final refit, but excludes data generation and metric computation. All methods were timed on the same hardware with controlled thread counts. The timing experiment is separate from the predictive benchmark and does not alter any reported performance result.

Standalone pipeline time (reference: s; high dimension: min): Nonboosting PU				
Method	Reference OS (s)	Reference TS (s)	$p = 500$ OS (min)	$p = 500$ TS (min)
Logistic probability correction	0.018 [0.0178, 0.0185]	0.0194 [0.019, 0.0195]	0.00153 [0.00147, 0.00156]	0.00185 [0.00169, 0.00204]
uPU logistic	2.7 [2.1, 3.2]	2.7 [1.8, 3.3]	0.0425 [0.0343, 0.0446]	0.0484 [0.0472, 0.0498]
nnPU logistic	2.6 [2.1, 3.1]	2.6 [1.7, 3.2]	0.0342 [0.0337, 0.0497]	0.0852 [0.0837, 0.0903]

Standalone pipeline time (reference: s; high dimension: min): Naive boosting				
Method	Reference OS (s)	Reference TS (s)	$p = 500$ OS (min)	$p = 500$ TS (min)
Naive AdaBoost	10.9 [10.8, 10.9]	10.8 [8.6, 10.9]	6.19 [6.17, 6.20]	6.26 [6.21, 6.27]
Naive GBM	9.5 [9.5, 9.5]	9.5 [9.5, 9.5]	7.61 [7.58, 7.62]	7.65 [7.65, 7.69]
Naive XGBoost	0.312 [0.311, 0.313]	0.315 [0.314, 0.318]	0.161 [0.159, 0.162]	0.166 [0.163, 0.166]
Naive LightGBM	0.451 [0.446, 0.46]	0.46 [0.459, 0.463]	0.244 [0.243, 0.245]	0.251 [0.25, 0.252]
Naive CatBoost	1.3 [1.3, 1.3]	1.3 [1.3, 1.3]	0.713 [0.71, 0.724]	0.758 [0.758, 0.761]

Standalone pipeline time (reference: s; high dimension: min): Probability-corrected boosting				
Method	Reference OS (s)	Reference TS (s)	$p = 500$ OS (min)	$p = 500$ TS (min)
Corrected AdaBoost	10.9 [10.8, 10.9]	10.8 [8.6, 10.9]	6.19 [6.17, 6.20]	6.26 [6.21, 6.27]
Corrected GBM	9.5 [9.5, 9.5]	9.5 [9.5, 9.5]	7.61 [7.58, 7.62]	7.65 [7.65, 7.69]
Corrected XGBoost	0.312 [0.311, 0.313]	0.315 [0.314, 0.318]	0.161 [0.159, 0.162]	0.166 [0.163, 0.166]
Corrected LightGBM	0.451 [0.446, 0.46]	0.46 [0.459, 0.463]	0.244 [0.243, 0.245]	0.251 [0.25, 0.252]
Corrected CatBoost	1.3 [1.3, 1.3]	1.3 [1.3, 1.3]	0.713 [0.71, 0.724]	0.758 [0.758, 0.761]

Standalone pipeline time (reference: s; high dimension: min): Reliable-negative boosting				
Method	Reference OS (s)	Reference TS (s)	$p = 500$ OS (min)	$p = 500$ TS (min)
RN AdaBoost	7.5 [7.2, 7.7]	7.9 [7.2, 8.1]	1.68 [1.60, 1.75]	1.68 [1.65, 1.95]
RN GBM	6.7 [6.6, 6.7]	8.1 [6.4, 8.5]	2.10 [2.00, 2.17]	2.01 [1.97, 2.42]
RN XGBoost	1.1 [1.1, 1.1]	1.2 [1.1, 1.2]	0.279 [0.271, 0.298]	0.287 [0.272, 0.326]
RN LightGBM	1.8 [1.7, 1.8]	1.8 [1.8, 1.9]	0.417 [0.416, 0.435]	0.421 [0.4, 0.504]
RN CatBoost	4.1 [4.1, 4.2]	4.2 [4.1, 4.2]	2.51 [2.49, 2.51]	2.60 [2.57, 2.61]

Standalone pipeline time (reference: s; high dimension: min): Cost-sensitive boosting				
Method	Reference OS (s)	Reference TS (s)	$p = 500$ OS (min)	$p = 500$ TS (min)
CS AdaBoost	28.4 [25.7, 28.4]	28.2 [26.1, 28.2]	18.67 [17.77, 18.68]	18.93 [18.86, 18.94]
CS GBM	31.6 [31.6, 31.7]	31.6 [31.6, 31.6]	24.99 [24.96, 25.00]	25.21 [25.18, 25.24]
CS XGBoost	1.1 [1.1, 1.1]	1.1 [1.1, 1.1]	0.537 [0.536, 0.542]	0.54 [0.537, 0.573]
CS LightGBM	1.6 [1.6, 1.6]	1.6 [1.6, 1.6]	0.887 [0.887, 0.888]	0.913 [0.903, 0.916]
CS CatBoost	4.9 [4.9, 4.9]	4.8 [4.8, 4.9]	2.69 [2.67, 2.69]	2.81 [2.78, 2.84]

Standalone pipeline time (reference: s; high dimension: min): Native PU boosting				
Method	Reference OS (s)	Reference TS (s)	$p = 500$ OS (min)	$p = 500$ TS (min)
AdaPU	2.6 [2.6, 2.8]	2.5 [2.4, 2.7]	2.14 [2.07, 2.19]	2.15 [2.10, 2.23]
PU gradient boosting	10.4 [10.3, 10.4]	10.0 [10.0, 10.0]	7.80 [6.17, 7.86]	5.46 [5.31, 6.10]

Standalone pipeline time (reference: s; high dimension: min): Supervised references				
Method	Reference OS (s)	Reference TS (s)	$p = 500$ OS (min)	$p = 500$ TS (min)
Supervised logistic regression	0.0534 [0.0533, 0.061]	0.0641 [0.0637, 0.065]	0.00255 [0.00226, 0.00255]	0.00299 [0.00256, 0.00302]
Supervised CatBoost	1.4 [1.3, 1.4]	1.3 [1.3, 1.3]	0.75 [0.747, 0.807]	0.761 [0.76, 0.763]

Incremental correction only (milliseconds)				
Method	Reference OS (ms)	Reference TS (ms)	$p = 500$ OS (ms)	$p = 500$ TS (ms)
Corrected AdaBoost	0.040 [0.038, 0.044]	0.259 [0.256, 0.270]	0.060 [0.057, 0.063]	0.358 [0.352, 0.375]
Corrected GBM	0.033 [0.032, 0.034]	0.224 [0.222, 0.226]	0.037 [0.037, 0.038]	0.249 [0.241, 0.256]
Corrected XGBoost	0.033 [0.033, 0.034]	0.214 [0.214, 0.217]	0.040 [0.040, 0.041]	0.222 [0.221, 0.227]
Corrected LightGBM	0.031 [0.031, 0.031]	0.215 [0.215, 0.219]	0.020 [0.020, 0.032]	0.217 [0.214, 0.222]
Corrected CatBoost	0.031 [0.030, 0.032]	0.214 [0.213, 0.214]	0.022 [0.021, 0.032]	0.218 [0.216, 0.219]

Note. OS: one-sample; TS: two-sample. Brackets give the 25th and 75th percentiles. Standalone reference times are seconds and high-dimensional times are minutes; incremental corrections in the final panel are milliseconds. Raw times are always retained in seconds. CPU: AMD EPYC 7543 32-Core Processor. Python 3.10.12. One library thread per worker; at most 20 case workers pinned to distinct physical cores on one exclusive node. Active concurrency decreases as cases finish. All methods share the same data within a case. Warm-up is discarded. Timers include internal scoring/threshold preparation and redirected fitting diagnostics. Corrected pipelines reuse the matching naive fit: standalone time adds the measured correction; the fit is not charged twice to the shared benchmark. Candidate counts and software versions are in the raw CSV. Candidate counts are validation candidates/checkpoints from the frozen selection hooks, not counts of separate trajectories; AdaPU can evaluate several stages from each beta trajectory. The two supervised references are separate comparators. The exploratory neural extension uses a distinct protocol and is excluded.